TEST CODE J

FORM TP 2019024

CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE EXAMINATION

PHYSICS

Paper 01 – General Proficiency

1 hour 15 minutes

17 JANUARY 2019 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer th
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Rea are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the same letter as the answer you have chosen. Look at the sample item below.

   Sample Item

   The SI unit of length is the

   (A) metre
   (B) second
   (C) newton
   (D) kilogram

   Sample Answer

   ● (B) (C) (D)

   The best answer to this item is "metre", so (A) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your ne
6. When you are told to begin, turn the page and work as quickly and as carefully a cannot answer an item, go on to the next one. You may return to that item later.
7. You may do any rough work in this booklet.
8. Figures are not necessarily drawn to scale.
9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO

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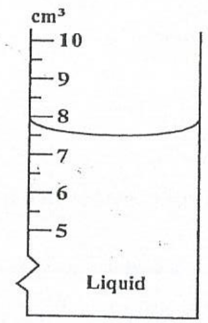
1. 0.0000462 N expressed in standard form is

(A) 4.62×10^{-5} N
(B) 4.62×10^{-4} N
(C) 462×10^{-4} N
(D) 4.62×10^{5} N

Item 2 refers to the following measuring cylinder which is used to determine the volume of a liquid.



2. The volume of the liquid is

(A) 7.0 cm^3
(B) 7.5 cm^3
(C) 7.8 cm^3
(D) 8.0 cm^3

3. One gram is equal to

(A) 10 milligrams
(B) 100 milligrams
(C) 1 000 milligrams
(D) 10 000 milligrams

4. Which of the following instruments is suitable for measuring the diameter of a human hair?

(A) Metre rule
(B) Tape measure
(C) Vernier caliper
(D) Micrometer screw gauge

5. Which of the following implements are designed to take advantage of a large moment provided by a relatively small force?

I. Crowbar
II. Claw hammer
III. Pair of tweezers

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

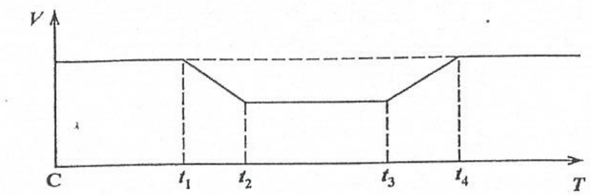
6. The mass of an astronaut is 70 kg when standing on the moon. When he returns to earth his approximate weight will be

(A) 70 kg
(B) 420 kg
(C) 70 N
(D) 700 N

7. Which of the following is a derived unit?

(A) K
(B) s
(C) m^3
(D) kg

Item 8 refers to the following diagram which shows a velocity/time graph for a moving object.



8. Which of the following statements about the object are true?

I. It returns to its starting point.
II. It has zero acceleration between times t_2 and t_3.
III. Its velocity at t_4 is the same as its initial velocity.

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

9. Two forces of 8 N and 10 N CANNOT give a resultant of

(A) 1 N
(B) 2 N
(C) 9 N
(D) 18 N

10. Which of the following features must be present in a stable, well-designed racing car?

(A) Low centre of gravity
(B) Narrow wheelbase
(C) Long front
(D) Sunroof

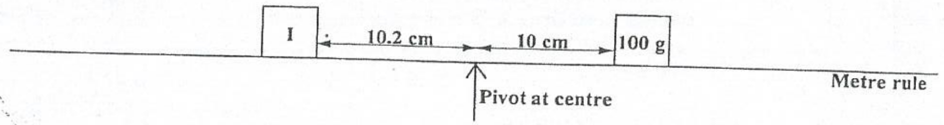
11. A student sets up a simple pendulum and finds that the period is 1.7 s. To obtain a period nearer to one second, he should

(A) use a lighter bob
(B) use a heavier bob
(C) lengthen the pendulum
(D) shorten the pendulum

12. When a ball is thrown vertically upwards and reaches its maximum height, it has

(A) maximum kinetic energy and maximum potential energy
(B) maximum kinetic energy and minimum potential energy
(C) maximum potential energy and minimum kinetic energy
(D) minimum kinetic energy and minimum potential energy

Item 13 refers to the following diagram.



13. The diagram above represents a 100 g mass which can be balanced by placing a mass at I. If th mass at I is to be used to balance the 100 g, it should be

(A) less than 100 g
(B) 100 g
(C) between 100 g and 199 g
(D) 200 g

14. What is the unit for the moment of a force?

(A) N
(B) N m
(C) N m^{-1}
(D) N m^{-2}

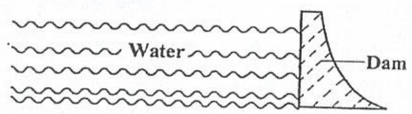
15. When two bodies collide, momentum is conserved. This means that the

(A) kinetic energy before impact is equal to that after impact
(B) momentum of each body is unchanged
(C) algebraic sum of the velocities before impact is equal to the sum of the velocities after impact
(D) total momentum of the bodies before impact is equal to the total momentum of the bodies after impact

16. A piece of string is tied onto a small stone and the stone is then totally immersed, in water. The tension in the string will be

(A) zero
(B) equal to the weight of the stone
(C) less than the weight of the stone
(D) more than the weight of the stone

Item 17 refers to the following diagran which shows a dam.



17. The pressure on the bottom of the dan depends on the

(A) depth of the water
(B) length of the reservoir
(C) volume of water held by the dam
(D) mass of water held back by the dam

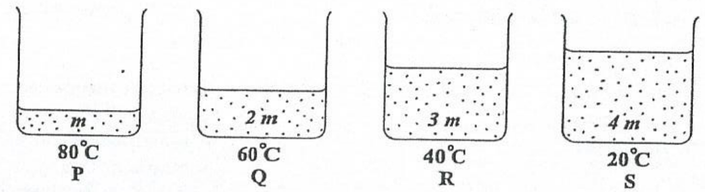
18. Which of the following scientists was responsible for arriving at the conclusion that measured amounts of electrical and mechanical energy can be converted to proportionate amounts of heat energy?

(A) Joule
(B) Coulomb
(C) Rumford
(D) Newton

19. The heat capacity of a substance is defined as the amount of heat energy

(A) the substance can hold
(B) 1 kg of the substance can hold
(C) required to change the substance to another state
(D) needed to raise the temperature of the substance by 1 degree

Item 20 refers to the following diagrams where P, Q, R and S are identical containers containing water of masses m, 2m, 3m and 4m respectively, at the temperatures indicated.



20. In which of the containers above is the average kinetic energy of the molecules greatest?

(A) P
(B) Q
(C) R
(D) S

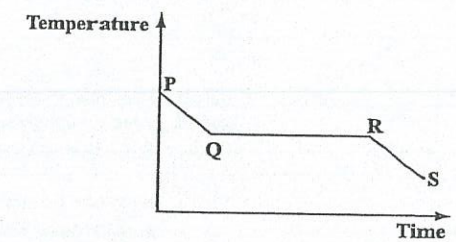
21. The energy required to change the state of a substance was determined to be E_H. If the mass of the substance was DOUBLED, the value of E_H will

(A) be halved
(B) be doubled
(C) be quadrupled
(D) remain constant

22. The specific latent heat of vaporization of water is the energy required to change 1 kg of water at

(A) 0 °C to ice at 0 °C
(B) 99.9 °C to steam at 100.1 °C
(C) 100 °C to steam at 100 °C
(D) 0 °C to steam at 100 °C

Item 23 refers to the following graph.



23. The graph above, arising from an experiment on change of phase, shows that solidification started at Q. During which of the stages is the substance in the liquid phase?

(A) At P only
(B) Between Q and R
(C) Between R and S
(D) Between P and Q

24. Which of the following remains unchanged with an increase in temperature?

(A) Density
(B) Mass
(C) Volume
(D) Relative density

25. Which of the following MOST likely accounts for the fact that pot handles are usually made of wood or plastic?

(A) Conduction
(B) Convection
(C) Evaporation
(D) Radiation

26. A detector of thermal energy is placed an equal distance in turn from each of four faces of a hollow metal cube full of hot water. The reading on the detector is GREATEST when the detector is turned towards the face which is

(A) painted shiny white
(B) painted dull black
(C) highly polished
(D) painted silver

27. Which of the following are reasons why a hot liquid, placed in a double-walled vacuum flask, retains its heat for a long time?

I. Evacuated space between the double walls reduces the loss of heat by conduction.
II. Silver inner walls reduce the loss of heat by radiation.
III. The silvered outer wall helps to absorb heat from the surroundings.

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

28. Which of the following is NOT an example of evaporation?

(A) A slice of bread left in the open air becomes dry.
(B) The cooling effect of sweating in animals.
(C) The rapid disappearance of ether if exposed to the air.
(D) A loaded copper wire put around a block of ice gradually cuts through the ice.

29. Two types of radiation, L and R, fall on a woman's left and right hands respectively. Her left hand feels hot but does not become suntanned; her right hand does not feel hot but it eventually becomes tanned. Radiations L and R are respectively

	L	R
(A)	visible light	x-rays
(B)	x-rays	visible light
(C)	ultraviolet	infrared
(D)	infrared	ultraviolet

30. The refractive index of a transparent medium with a critical angle, c, for light travelling from the medium to air is

(A) $\frac{1}{c}$

(B) $\frac{90°}{\sin c}$

(C) $\frac{\sin 90°}{\sin c}$

(D) $\sin c$

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31. A ray of light leaves air and enters glass of refractive index 1.6. The angle of refraction is 27°. What is the sine of the angle of incidence?

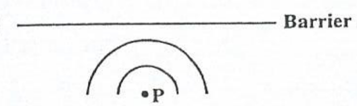
(A) $1.6 + \sin 27°$

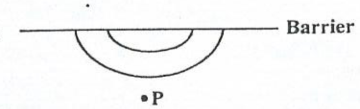
(B) $\frac{1.6}{\sin 27°}$

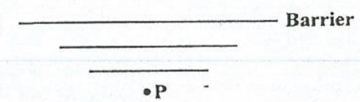
(C) $1.6 \sin 27°$

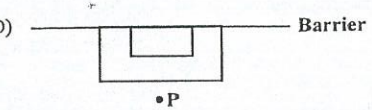
(D) $\frac{\sin 27°}{1.6}$

32. A pencil point is dipped into water at Point P in front of a straight barrier. Which of the following will the REFLECTED wave look like?

(A) 

(B) 

(C) 

(D) 

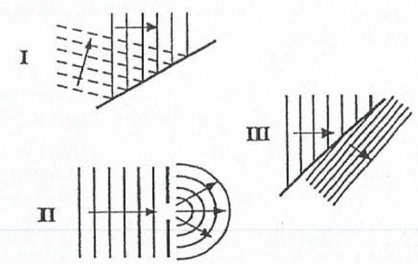
Item 33 refers to the following table which lists the refractive indices for light in four different media.

Medium	Refractive Index
Air	1.0
Ice	1.3
Perspex	1.5
Diamond	2.4

33. In which medium would the light waves have the slowest speed?

(A) Air
(B) Ice
(C) Perspex
(D) Diamond

34. Which of the following diagrams represent(s) diffraction of water waves in a ripple tank?



(A) I only
(B) II only
(C) I and II only
(D) II and III only

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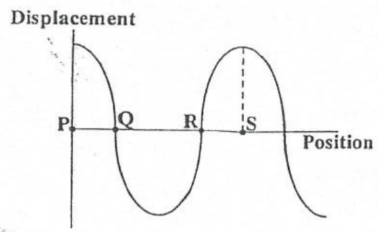
Item 35 refers to the following diagram which shows a transverse wave at a particular instant.



35. The wavelength of the wave is equal to the distance

(A) PQ
(B) PR
(C) PS
(D) QR

36. Which of the following waves travel(s) only longitudinally?

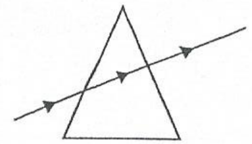
I. Sound waves
II. Radio waves
III. Water waves

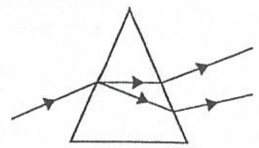
(A) I only
(B) II only
(C) II and III only
(D) I, II and III

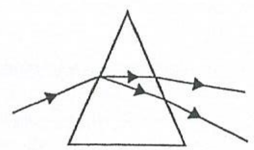
37. The waves that reach a listener from a radio are

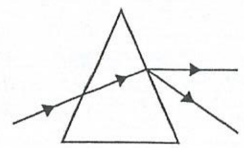
(A) electromagnetic
(B) transverse and stationary
(C) longitudinal and stationary
(D) longitudinal and progressive

38. Which of the following diagrams BEST represents the passage of a beam of white light through a triangular glass prism?

(A)


(B)


(C)


(D)


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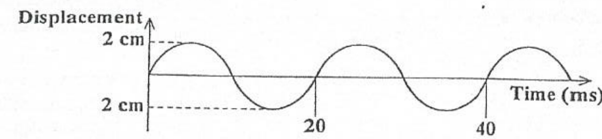
Item 39 refers to the following diagram which shows an instantaneous profile of a wave travelling across a water surface.



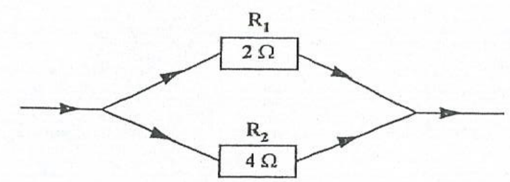
39. From the information given, the frequency is

(A) 20 Hz
(B) 25 Hz
(C) 50 Hz
(D) 100 Hz

40. The note from a drum is louder when it is struck harder because the sound waves produced have a greater

(A) amplitude
(B) wavelength
(C) frequency
(D) velocity

41. What is the equivalent resistance of the two resistors in the following diagram?



(A) 0.5 Ω
(B) 1.3 Ω
(C) 6 Ω
(D) 8 Ω

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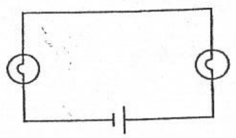
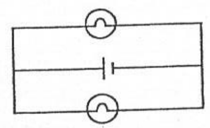
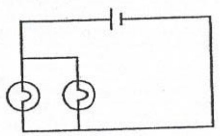
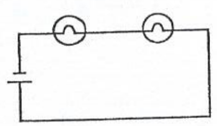
Item 42 refers to the following diagram which shows an electrical circuit with a cell and two filament bulbs.

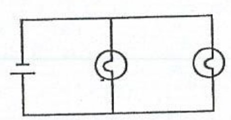


42. Which of the following circuits are electrically the same as the circuit above?

(A)


(B)


(C)


(D)


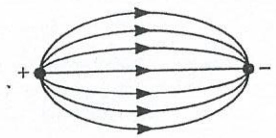
43. The SI unit of charge is the

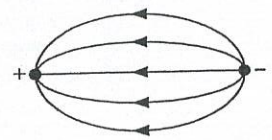
(A) ohm
(B) volt
(C) ampere
(D) coulomb

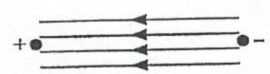
44. Which of the following is NOT one of the ways in which the strength of the magnetic field near a solenoid (long coil) carrying a current can be increased?

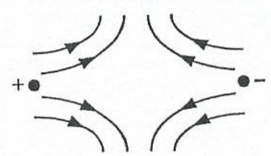
(A) Increasing the resistance of the coil
(B) Increasing the current in the coil
(C) Increasing the number of turns per unit length of the coil
(D) Placing a soft iron core inside the coil

45. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?

(A)


(B)


(C)


(D)


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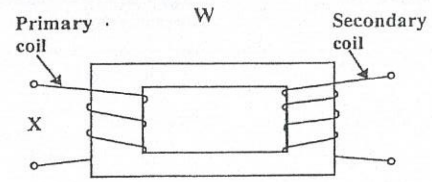
Item 46 refers to the following diagram.



46. Appropriate labels for W and X respectively would be

(A) step-up transformer, d.c. input
(B) step-up transformer, a.c. input
(C) step-down transformer, d.c. input
(D) step-down transformer, a.c. input

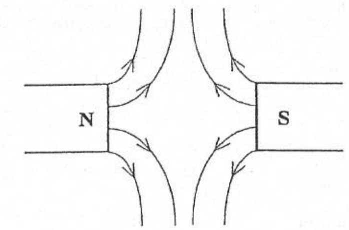
47. An ideal transformer has 1200 turns on the primary and 600 turns on the secondary. If the voltage across the secondary is 20 V, the voltage across the primary is

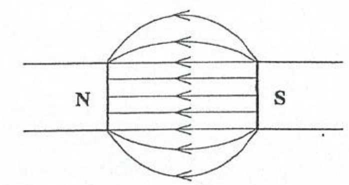
(A) 5 V
(B) 10 V
(C) 20 V
(D) 40 V

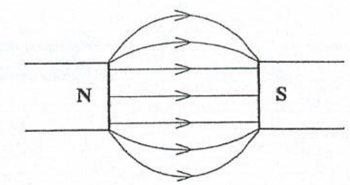
48. A glass rod is rubbed with a piece of silk and is positively charged. The glass rod became charged by

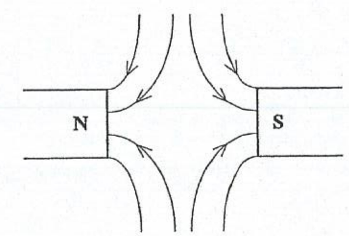
(A) losing protons
(B) losing electrons
(C) gaining protons
(D) gaining electrons

49. Which of the following diagrams represents the magnetic field which exists between two opposite magnetic poles?

(A)


(B)


(C)


(D)


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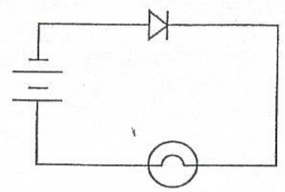
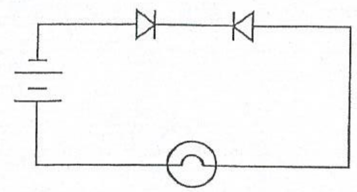
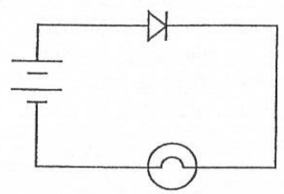
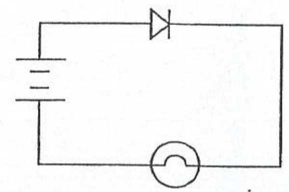
50. In which of the following circuits will the lamp light up?

(A) 

(C) 

(B) 

(D) 

51. Given the following truth table with inputs A and B and output C, which logic gate does it describe?

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

(A) NAND
(B) NOR
(C) AND
(D) OR

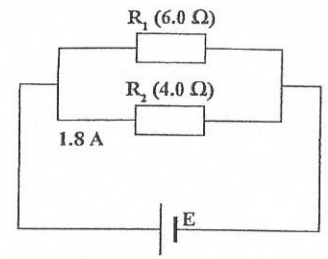
52. An electromagnet consists of insulated wire wrapped around an iron core. It works because

(A) iron is always magnetized
(B) iron is a good electrical conductor
(C) a magnetic field is produced inside the coil
(D) an electric field is produced inside the coil

53. When an excessive current passes through a fuse, which of the following is the sequence of events?

(A) Wire gets hot → current is cut off → wire melts
(B) Wire gets hot → wire melts → current is cut off
(C) Wire melts → current is cut off → wire gets hot
(D) Wire melts → wire gets hot → current is cut off

Item 54 refers to the following diagram which shows two resistors, R_1 of 6.0 Ω and R_2 of 4.0 Ω, in parallel.



54. What is the current through R_1 if the current through R_2 is 1.8 A?

(A) 1.2 A
(B) 1.8 A
(C) 2.7 A
(D) 3.0 A

55. An ammeter has a very low resistance so that it can be placed in

(A) parallel with a component and not affect the circuit
(B) series with a component and not affect the circuit
(C) parallel with a component and not heat up
(D) series with a component and not heat up

56. The nuclide $^{234}_{90}Th$ contains

(A) 90 protons and 234 neutrons
(B) 235 protons and 90 neutrons
(C) 90 protons and 144 neutrons
(D) 144 protons and 90 neutrons

57. Madame Curie's work in scientific research was chiefly concerned with

(A) radioactivity
(B) atomic structure
(C) nuclear structure
(D) energy from atomic nuclei

58. Which of the following describes two properties of an α-particle?

(A) No charge, very penetrating
(B) Positive charge, very penetrating
(C) Negative charge, not very penetrating
(D) Positive charge, not very penetrating

59. Sodium-24 decays into Magnesium-24 with the emission of a β-particle and can be represented by the following equation.

$$^{24}_{11}\text{Na} \rightarrow {}^{24}_{12}\text{Mg} + \boxed{}$$

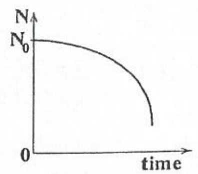
Which of the following options should be placed in the box to complete the equation?

(A) $^{0}_{1}e$
(B) $^{0}_{-1}e$
(C) $^{4}_{2}\text{He}$
(D) $^{0}_{1}\text{He}$

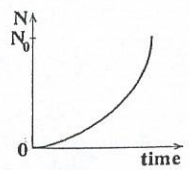
60. The number of radioactive nuclei present in a sample of the time $t = 0$ is N_0.

Which of the following graphs BEST represents the variation with time of the number, N, of undecayed nuclei present?

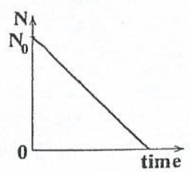
(A)



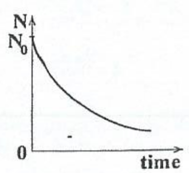
(B)



(C)



(D)



END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.